

# Proposed Solution

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| Date          | 17 September 2026 |
| Team ID       | SWTID-2026-9627   |
| Project Name  | AI Fit Track API  |
| Maximum Marks | 2 Marks           |

## Proposed Solution

| Element                      | Description                                                                                                                                                                                                                                                                                                        |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Project Title                | AI Fit Track - AI-Powered Workout Tracking & Fitness Coaching Backend                                                                                                                                                                                                                                              |
| Objective                    | Build a secure, RESTful Node.js/Express backend that lets an authenticated user manage their workout history and receive AI-generated (Google Gemini) workout recommendations and fitness insights derived from that history.                                                                                      |
| Scope                        | User registration/login (JWT + bcrypt), full CRUD and search on Workout records, and two Gemini-backed AI endpoints (recommendation and insights). A client/ folder is scaffolded for a future UI but a dedicated frontend is out of scope for this cycle - the API is validated through Thunder Client / Postman. |
| Problem it Solves            | Removes fragmented manual logging (notebooks/spreadsheets), removes manual-entry data errors via schema validation, and adds a personalized AI guidance layer that no manual method can offer.                                                                                                                     |
| Proposed Solution / Approach | A modular MVC-style Express application ( config / controllers / middleware / models / routes / services ) backed by MongoDB via Mongoose, JWT-protected routes, and a dedicated geminiService.js that isolates all Gemini AI calls from the rest of the business logic.                                           |
| Resource Requirements        | Node.js v16+, Express.js, MongoDB (local instance, mongodb://localhost:27017/aifittrack ), Mongoose, jsonwebtoken , bcryptjs , Google Gemini API key, VS Code + Thunder Client/Postman, Git & GitHub, 5-member team over the 15-29 September 2026 cycle.                                                           |